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United States Patent	12410943
Kind Code	B2
Date of Patent	September 09, 2025
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Heating, ventilation, and air conditioning system with tiered multi-level base pan

Abstract

A heating, ventilation, and air conditioning (HVAC) system includes one or more compressors, one or more outdoor heat exchangers, and one or more indoor heat exchangers. The HVAC system also includes a base pan on which the one or more compressors are mounted. The base pan includes a base level and one or more tier levels higher than the base level. The distance from any point on a perimeter of the first-tier level to a nearest point on a perimeter of the base level is based on the smallest bolt circle diameter of the one or more compressors mounted on the base pan. Also, the perimeter lengths of the tier levels are based on a percentage of the base level perimeter length.

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Appl. No.: 18/733541

Filed: June 04, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240318864 A1	Sep. 26, 2024

Related U.S. Application Data

continuation parent-doc US 17373979 20210713 PENDING child-doc US 18733541

Publication Classification

Int. Cl.: F24F13/32 (20060101); F24F13/20 (20060101); F24F13/22 (20060101); F24F13/30 (20060101)

U.S. Cl.:

CPC F24F13/32 (20130101); F24F13/20 (20130101); F24F13/22 (20130101); F24F13/222 (20130101); F24F13/30 (20130101); F24F2013/202 (20130101)

Field of Classification Search

CPC: F24F (13/32); F24F (13/20); F24F (13/22); F24F (13/222); F24F (13/30); F24F (2013/202); F24F (1/10); F24F (1/36); F24F (1/56)

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Patent No.	Application Date	Country	CPC
200178858	12/1999	KR	N/A

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Background/Summary

BACKGROUND

(1) This section is intended to provide relevant background information to facilitate a better understanding of the various aspects of the described embodiments. Accordingly, these statements are to be read in this light and not as admissions of prior art.

(2) Typically, one or more refrigerant compressors of a heating, ventilation, and air conditioning (HVAC) system is mounted on a base pan designed to support these one or more compressors. Base pans can potentially be insufficiently structurally supportive such that the base pan sags, especially with the compressor and other HVAC system components such as accumulator mounted on top of the base pan. This sag can present the problem of collecting water, which over time can damage the base pan, as well as HVAC system components attached to the base pan. Further, some base pan designs suffer from vibration fatigue from the compressor or compressors during operation, which can be increased if the vibration frequency of the compressor happens to be a resonant frequency of the base pan. Vibration of the base pan from operation of the compressor or compressors can also damage the base pan over time. Additionally, during shipment, the compressor or other components of the HVAC system tends to vibrate more violently than during normal operation of the air conditioning unit. Base pans that allow excess movement and vibration during shipping may result in damage to the HVAC system components such as compressors and accumulators, and the base pan. Such vibrations during operation or shipping can also lead to failure in the piping system and other system components due to the high stresses generated by induced structural loads and displacements. HVAC systems can be shipped from the factory with the compressor “tied down”

against the base pan to prevent damage while the unit is in transit. However, vibration of a “tied down” compressor may still lead to damage to the base pan. In addition, tying down the compressor requires additional time when preparing the air conditioning unit for shipment and then when installing the unit to tighten and loosen the compressor, as needed. Additional costs associated with these extra steps are also incurred. Furthermore, excessive vibration may cause undesired noise propagating into the conditioned environment and causing environmental disturbance.

(3) An HVAC system with a robust base pan is needed that is cost effective and can be efficiently assembled for shipment and operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the HVAC system with a tiered multi-level base pan are described with reference to the following figures. The same or sequentially similar numbers are used throughout the figures to reference like features and components. The features depicted in the figures are not necessarily shown to scale. Certain features of the embodiments may be shown exaggerated in scale or in somewhat schematic form, and some details of elements may not be shown in the interest of clarity and conciseness.

(2) FIGS. 1A and 1B are a perspective and cross-section views of a heating ventilation and air conditioning (HVAC) system, according to one or more embodiments;

(3) FIGS. 2A and 2B are a top and cross section-views of an embodiment of a base pan for use in an HVAC system; and

(4) FIG. 3 is a top view of another embodiment of a base pan for use in an HVAC system.

DETAILED DESCRIPTION

(5) The present disclosure describes a heating, ventilation, and air conditioning (HVAC) system with one or more compressors mounted onto a tiered multi-level base pan. Although the general description of this invention is referenced to the HVAC system compressors mounted on the base pan, other refrigerant system components including but not limited to accumulators, receivers, charge compensators, flow control devices, air movers, pumps, and filter driers can be structurally attached to the base pan and may require similar treatment. The design of the base pan is configured such that the vertical forces from the compressor(s) mounted on the base pan are evenly distributed across the base pan and transmitted from the center of the base pan outward to the perimeter of the base pan. The geometry of each tiered level is optimized to minimize the amount of flat, unsupported material, which results in minimized deflection of the base pan under load from the mounted compressor, as well as allowing any accumulated water to drain towards the unit periphery and be discharged outside of the unit boundaries. The tiered multi-level design includes a base level and at least one tier level higher than the base level. The base pan may also be supported by one or more base level support structures, typically near or along a perimeter of the base level. A distance from a perimeter of the first tier level to the perimeter of the base level or a base level support structure is based on the smallest bolt circle diameter of the compressor(s). Further, the tier level perimeter lengths are based on a percentage of the base level perimeter length. Further, the base pan includes one or more compressor structural supports and the distance from a center of one of the compressors to the closest compressor structural support is equal to two times the smallest bolt circle diameter or less.

(6) The geometry of the base level and each tier level is optimized to balance the forces from the compressor(s), thus minimizing side-to-side motion of the compressor(s) and keeping the motion of the compressor(s) limited and linear in the vertical direction. The linear motion means all compressor feet move in unison, spreading out the load of each compressor evenly and minimizing

stress concentrations in the base pan. The vertical motion also minimizes torsional stresses in associated connection tubing, since the combination of bending and torsional stresses accelerates fatigue failure in the connecting refrigerant tubing. The resulting structure also dampens harmonics that may amplify during transit of the system. The tiered design also ensures that any water on the base pan surface drains toward the outside of the base pan. As noted above, analogous conclusions can be drawn regarding other HVAC system components, such as accumulators, receivers, charge compensators, flow control devices, air movers, pumps, and filter driers structurally attached to the base pan and requiring similar treatment.

(7) Turning now to the figures, FIGS. 1A and 1B show an external appearance of an HVAC system **100** when seen from obliquely above the HVAC system **100** as well as a cross-section view of the HVAC system **100** from a top view perspective. Although not shown, it should be appreciated that the HVAC system **100** includes additional panel covers for covering and protecting the equipment of the HVAC system **100**. The example HVAC system **100** shown is a so-called “light” commercial package unit and shall be described in terms of a cooling operation, although it should be appreciated that the HVAC system **100** could also be used for heating and can be representing residential packaged, residential split, light commercial split, or commercial applied applications. The HVAC system **100** includes both an “outdoor” section **SP1** and an “indoor” section **SP2** mounted on a common frame **102**.

(8) The outdoor section **SP1** includes one or more compressors **104** mounted on a base pan **106**. As noted above, the outdoor section **SP1** may include other HVAC system components, such as accumulators, receivers, charge compensators, flow control devices, air movers, pumps, and filter driers structurally attached to the base pan (not shown). Also included are one or more outdoor heat exchangers **108** and outdoor fans **110** that move air into the outdoor section **SP1** across the outdoor heat exchanger **108** and to the outside of the HVAC system **100**. FIG. 1A is shown without the additional outdoor heat exchanger **108** and outdoor fan **110** to be able to view additional structure and equipment. However, FIG. 1B shows the additional outdoor heat exchanger **108** and it should be appreciated that an additional outdoor fan **110** would be included as well. The outdoor fans **110** may be any suitable type of fan, for example, a propeller fan. The outdoor heat exchangers **108** may include a plurality of heat-transfer tubes (not shown) in which a refrigerant flows and a plurality of heat-transfer fins (not shown) in which air flows between gaps thereof. The plurality of heat-transfer tubes may be arranged in an up-down direction (hereunder may be referred to as “row direction”), and each heat-transfer tube may extend in a direction substantially orthogonal to the up-down direction (in a substantially horizontal direction). At an end portion of the outdoor heat exchangers **108**, for example, the heat-transfer tubes are connected to each other by being bent into a U-shape or by using a U-shaped return bends so that the flow of a refrigerant from a certain column to another column and/or a certain row to another row is turned back. The plurality of heat-transfer fins that extend so as to be long in the up-down direction are arranged side by side in a direction in which the heat-transfer tubes extend with a predetermined interval between the plurality of heat-transfer fins. The plurality of heat-transfer fins and the plurality of heat-transfer tubes are assembled to each other so that each heat-transfer fin extends through the plurality of heat-transfer tubes. The plurality of heat-transfer fins are also disposed in a plurality of columns.

(9) Due to the structure of the outdoor heat exchangers **108**, a flow path of outdoor air that enters the outdoor space **SP1** passes through the outdoor heat exchanger **108s**, where the outdoor air exchanges heat with a refrigerant that flows in the outdoor heat exchangers **108**. Air after the heat exchange in the outdoor heat exchanger **108** is discharged to the outside of the outdoor space **SP1** by the outdoor fans **110**.

(10) The outdoor space **SP1** and the indoor space **SP2** are separated by a partition plate **112**. Outdoor air flows to the outdoor space **SP1** and indoor air flows to the indoor space **SP2**. By separating the outdoor space **SP1** and the indoor space **SP2** by the partition plate **112**, the airflow bypass between the outdoor space **SP1** and the indoor space **SP2** is blocked. Therefore, in an

ordinary state, the indoor air and the outdoor air do not mix and do not communicate with each other within or via the HVAC system **100**. It has to be noted, that there exist the airside economizers that allow mixing indoor and outdoor air, however there are not discussed in relation to this invention.

(11) Although not shown, the indoor section SP2 includes an expansion valve and a combustion heat exchanger located within a cabinet of the indoor section SP2. The expansion valve may alternatively be located in the outdoor section SP1. The indoor section SP2 also includes an indoor heat exchanger **116** and an indoor blower **118**, which may be, for example, a centrifugal fan. The indoor heat exchanger **116** may also include a plurality of heat-transfer tubes in which a refrigerant flows, and a plurality of heat-transfer fins in which air flows between gaps thereof. The plurality of heat-transfer tubes may be arranged in an up-down direction (row direction), and each heat-transfer tube may extend in a direction substantially orthogonal to the up-down direction (in the second embodiment, in a left-right direction). At an end portion of the indoor heat exchanger **116**, for example, the heat-transfer tubes are connected to each other by being bent into a U-shape or by using a U-shaped return bends so that the flow of a refrigerant from a certain column to another column and/or a certain row to another row is turned back. The plurality of heat-transfer fins and the plurality of heat-transfer tubes may be assembled so that each heat-transfer fin extends through the plurality of heat-transfer tubes.

(12) The indoor heat exchanger **116** divides the indoor space SP2 into a space on an upstream side with respect to the indoor heat exchanger **116** and a space on a downstream side with respect to the indoor heat exchanger **116**. All air that flows to the downstream side from the upstream side with respect to the indoor heat exchanger **116** passes through the indoor heat exchanger **116**. The indoor blower **118** is disposed in the space on the downstream side with respect to the indoor heat exchanger **116** and causes an airflow that passes through the indoor heat exchanger **116** to be generated. Although not shown, supply air and return air ducts are connected to the indoor space SP2 through a bottom plate **114** in the bottom of the HVAC system **100**. The blower **118** is disposed above a supply air opening **120** in the bottom plate **114** for providing supply air to the indoor space being cooled. A return air opening **122** in the bottom plate **114** provides return air from the indoor space being cooled to flow through the indoor heat exchanger **116** and the indoor blower **118**.

(13) The HVAC system **100** also includes a refrigerant circuit that includes the indoor heat exchanger **116** and the outdoor heat exchangers **108**. In the refrigerant circuit, a refrigerant circulates between the indoor heat exchanger **116** and the outdoor heat exchangers **108**. In the refrigerant circuit, when, in a cooling operation or a heating operation, a vapor compression refrigeration cycle is performed, heat is exchanged at the indoor heat exchanger **116** and the outdoor heat exchangers **108**. The refrigerant circuit includes the compressors **104**, the outdoor heat exchangers **108**, the expansion valve, and the indoor heat exchanger **116**. At the time of the cooling operation, a refrigerant is compressed by the compressors **104** and is sent to the outdoor heat exchangers **108**. The refrigerant dissipates heat to outdoor air at the outdoor heat exchangers **108** and is sent to the expansion valve. At the expansion valve, the refrigerant expands and its pressure and temperature are reduced, and is then sent to the indoor heat exchanger **116**. A refrigerant having a low temperature and a low pressure sent from the expansion valve exchanges heat at the indoor heat exchanger **116**, and absorbs heat from indoor air. The air cooled by having its heat taken away at the indoor heat exchanger **116** is supplied to the indoor space being cooled. The refrigerant after the heat exchange at the indoor heat exchanger **116** is then sucked into the compressors **104** to repeat the cycle.

(14) The equipment of the refrigerant circuit, and thus flow of the refrigerant through the circuit may be controlled by a main controller that controls the HVAC system **100**. The main controller may also be configured to be capable of communicating with a remote controller. A user can send, for example, a set values of indoor temperatures of rooms in the indoor space being cooled to the

main controller from the remote controller. For controlling the HVAC system **100**, a plurality of temperature sensors for measuring the temperature of a refrigerant at each portion of the refrigerant circuit and/or a pressure sensor that measures the pressure of each portion and a temperature sensor for measuring the air temperature of each location may be provided.

(15) The main controller performs at least on/off control of the compressors **104**, on/off control of the outdoor fans **110**, and on/off control of the indoor blower **118**. When any or all of the compressors **104**, the outdoor fans **110**, and the indoor blower **118** include a motor of a type whose number of rotations is changeable, the main controller may be configured to be capable of controlling the number of rotations of the motor or motors whose number of rotations is changeable among the motors of the compressors **104**, the outdoor fans **110**, and the indoor blower **118**. In this case, the main controller can control the circulation amount of the refrigerant that flows through the refrigerant circuit by changing the number of rotations of the motor of the compressors **104**. The main controller can change the flow rate of outdoor air that flows between the heat-transfer fins of the outdoor heat exchangers **108** by changing the number of rotations of the motor of the outdoor fans **110**. The main controller can change the flow rate of indoor air that flows between the heat-transfer fins of the indoor heat exchanger **116** by changing the number of rotations of the motor of the indoor blower **118**.

(16) The main controller may be realized by, for example, a computer. The computer that constitutes the main controller includes a control calculation device and a storage device. For the control calculation device, a processor such as a CPU or a GPU may be used. The control calculation device reads a program that is stored in the storage device and performs a predetermined image processing operation and a computing processing operation in accordance with the program. Further, the control calculation device writes a calculated result to the storage device and reads information stored in the storage device in accordance with the program. However, the main controller may be formed by using an integrated circuit (IC) that can perform control similar to the control that is performed by using a CPU and a memory. Here, IC includes, for example, LSI (large-scale integrated circuit), ASIC (application-specific integrated circuit), a gate array, and FPGA (field programmable gate array).

(17) FIGS. **1A** and **1B** illustrate the compressors **104** being a vertically oriented, generally cylindrical body supported by support structures **154**. The support structures **154** include a plurality of outwardly projecting “feet”, each of which has an opening **158** formed therein for attaching to the base pan **106** through appropriate fasteners or structurally supported interfaces.

(18) The base pan **106** may be manufactured from a sheet metal material. Alternatively, the base pan **106** be any material suitable for supporting the HVAC system **100** and may be manufactured by any suitable process, such as press-forming or additive manufacturing. The portion of the base pan **106** to which the compressor **104** is mounted, includes a plurality of holes complementary to the openings **158** of the feet of the support structure **154**. A fastener may be inserted through each hole in the base pan **106** and through an adjacent opening **158** in the support structure **154** to couple the compressor **104** to the base pan **106**. Additionally, although the base pan **106** and the bottom plate **114** are shown as separate, the base pan and the bottom plate may be integrally formed together to support equipment in both the outdoor space **SP1** and the indoor space **SP2**.

(19) FIG. **2A** shows a top view of an embodiment of a tiered multi-level base pan **206** for use with an HVAC system such as the HVAC system **100** discussed above. FIG. **2B** shows a cross-section view of the tiered multi-level base pan **206** taken through plane A-A. As shown, the base pan **206** includes a base level **260** that forms the base structure of the base pan **206**. Although shown shaped as a rectangle, the base level **260** could be shaped in any suitable form. The base level **260** also includes a wall **262** extending downward that surrounds the perimeter of the base level **260** and helps support the load of the base pan **206** itself as well as the compressor and other HVAC system components when mounted onto the base pan **206**. Underneath the base pan **206** are base level support structures **263**. Although two base level support structures **263** are shown, it should be

appreciated the base pan may only include one or more than two base level support structures **263**. During shipment of an HVAC system and in final installation, the base pan **206** may be placed on and supported by the base level support structures **263** underneath the base pan **206**. As an example, the base level support structures may include rails for supporting the base pan **206**.

(20) The base pan **206** includes at least one tier level **264** higher than the base level **260** and connected by a wall **266**. The wall **266** need not be completely perpendicular to the flat surfaces of the base level **260** and the tier level **264** but instead may be formed at an angle to perpendicular other than flat. Additionally, the transition from flat surface to wall and back to flat surface may be any suitable transition, such as a curved transition. As shown, the base pan **206** includes optional additional tier layers **264**, each increasing in height from the previous tier layer **264** and connected to the previous tier layer **264** by a wall **266**.

(21) The portion of the base pan **206** to which the compressor is mounted includes a plurality of holes **210** for mounting one or more compressors on top of the base pan **206**. The holes **210** are shown extending through compressor structural supports **212** raised from the highest tier level **264**. While the compressor structural supports **212** are shown as circular in shape, the structural supports **212** may be of any suitable shape and also may be sized such that more than one hole **210** extends through a compressor structural support **212**. The compressor(s) may also be mounted using less than all of the holes **210**. Each compressor that will be mounted to the base pan **206** has a "bolt circle," which is a circle that surrounds all of the holes **210** needed to mount a particular compressor. The holes **210** may be placed to accommodate compressors with different size bolt circles such that a configuration of the holes **210** may be used to mount compressors with different mounting locations without having to redesign the base pan **206**.

(22) As mentioned above, the design of the base pan **206** is configured such that the vertical forces from the compressor(s) mounted on the base pan **206** are evenly distributed across the base pan **206** and transmitted from the center of the base pan **206** outward to the perimeter **265** of the base pan **206**. The geometry of the base level **260** and each tier level **264** is optimized to minimize the amount of flat, unsupported material, which results in minimized deflection of the base pan **206** under load from the mounted compressor.

(23) One aspect of the design of the base pan **206** is that a distance B from any point on a perimeter of a tier level **264** to a nearest point on a perimeter **265** of the base level **260** or a base level support structure is based on a diameter of the bolt circle of the compressor to be mounted to the base pan **206**. For example, the distance from any point on the perimeter of the first tier level **264** to the nearest point on the perimeter **265** of the base level **260** is equal to two times the diameter of the bolt circle or less. If the bolt circle were 10.61 inches (0.269 m), then the distance would be 21.22 inches (0.564 m) or less. For another example, this distance may be the diameter of the compressor bolt circle or less. For a bolt circle of 10.61 inches (0.269 m), then the distance would be 10.61 inches (0.269 m) or less. If more than one compressor is to be mounted on the base pan **206**, then the distance from any point on the perimeter of the first tier level **264** to the nearest point on the perimeter **265** of the base level **260** is based on the smallest of the diameters of the bolt circles of the compressors. If there is more than one tier level **264**, e.g., second and third tier levels **264**, a distance from any point on the perimeter of any tier level **264** and the nearest point on the perimeter of the adjacent tier level **264**, for example distance C, is the diameter of the bolt circle of the compressor or less. For a bolt circle of 10.61 inches (0.269 m), then the distance would be 10.61 inches (0.269 m) or less. For example, this distance may be half of the diameter of the compressor bolt circle or less. For a bolt circle of 10.61 inches (0.269 m), then the distance would be 5.31 inches (0.135 m) or less.

(24) Another aspect of the design of the base pan **206** is that the length of the perimeter of any tier level **264** is based on a percentage of the length of the perimeter **265** of the base level **260**. For example, the length of the perimeters of the tier levels **264** range from 15% to 70% of the length of the perimeter **265** of the base level **260**. For example, the length of the perimeter of the tier levels

264 may be 60-70%, 45-55%, and 30-40% of the length of the perimeter **265** of the base level **260**. For example, the length of the perimeter of the first tier level **264** may be 60% to 70% of the length of the perimeter **265** of the base level **260**. The percentage of the length of the base level **260** for the length of the perimeter of any tier level **264** may also be selected based on the particular tier level number, e.g., tier level one, tier level two, etc. and expressed by:

$$(25) \text{ \%oflengthofbaselevelperimeter} = 1 - (0.2 + .15 * (\text{tierlevelnumber})) \quad (\text{Eq. 1})$$

and includes up to and including $\pm 5\%$ of the calculated percentage. This concept may be applied even if more or fewer tier levels **264** are used than shown in FIGS. 2A and 2B.

(26) Another aspect of the design of the base pan **206** is that the base pan **206** includes one or more compressor structural supports **212** and the distance from a center of the compressor to the closest compressor structural support **212** is equal to or less than two times the diameter of the bolt circle of the compressor. For example, this distance may be equal to or less than the diameter of the bolt circle of the compressor. If more than one compressor is to be mounted on the base pan **206**, then the distance from a center of either compressor to the closest compressor structural support **212** is based on the smallest of the diameters of the bolt circles of the compressors.

(27) FIG. 3 shows a top view of an alternative embodiment of a tiered base pan **306** for use with an HVAC system such as the HVAC system **100** discussed above. The base pan **306** is designed according to the same principles used in designing the base pan **206** discussed above. However, the base pan **306** includes a base level **360** and only two tier levels **364** compared to three tier levels shown in the base pan **206**. The base pan **306** further includes compressor structural supports **312** with holes **310** for mounting one or more compressors. However, some of the compressor structural supports **312** include more than one hole **310** each rather than just one hole each as shown in the base pan **206**.

(28) The embodiments of the base pans **206** and **306** shown in FIGS. 2A, 2B, and 3 show that there are multiple possible designs of the tiered based pan in accordance with the listed design principles. Each designed base pan includes a base level and at least one tier level that are optimized to balance the forces from the compressor, thus minimizing side-to-side motion of the compressor and keeping the motion of the compressor linear in the vertical direction. The vertical linear motion minimizes stress concentrations in the base pan and minimizes torsional stresses in associated connection tubing. The resulting structure also dampens harmonics that may amplify during transit of the system. The tiered multi-level design also ensures that any water on the base pan surface drains toward the outside of the base pan. It has to be noted that the base pan design can consist of a single piece construction encompassing both outdoor section SP1 and indoor section SP2.

(29) Alternatively, the base pan design can be a dedicated outdoor section SP1 base pan.

(30) Numbered examples of the above embodiments include: 1. A heating, ventilation, and air conditioning (HVAC) system comprising: one or more compressors; one or more outdoor heat exchangers; one or more indoor heat exchangers; and a base pan on which the one or more compressors are mounted and comprising a base level and one or more tier levels higher than the base level, wherein a distance from any point on a perimeter of the first tier level to a nearest point on a perimeter of the base level is based on the smallest bolt circle diameter of the one or more compressors and the tier level perimeter lengths are based on a percentage of the base level perimeter length. 2. The system of example 1, wherein the distance from any point on the perimeter of the first tier level to the nearest point on the perimeter of the base level is equal to two times the smallest bolt circle diameter or less. 3. The system of example 2, wherein the distance from any point on the perimeter of the first tier level to the nearest point on the perimeter of the base level is equal to the smallest bolt circle diameter or less. 4. The system of example 1, wherein the base pan further comprises more than one tier level and the distance from any point on the perimeter of each tier level and the nearest point on the perimeter of the adjacent tier level is based on the smallest bolt circle diameter. 5. The system of example 4, wherein the distance from any point on the

perimeter of each tier level and the nearest point on the perimeter of the adjacent tier level is equal to the smallest bolt circle of the compressor or less. 6. The system of example 4, wherein the distance from any point on the perimeter of each tier level and the nearest point on the perimeter of the adjacent tier level is equal to half the smallest bolt circle of the compressor or less. 7. The system of example 1, wherein the base pan further comprises more than one tier level and each tier level perimeter length is based on a percentage of the base level perimeter length. 8. The system of example 7, wherein the percentage of the base level perimeter length for each tier level perimeter length is selected based on the tier level number. 9. The system of example 8, wherein the percentage of the base level perimeter length of each tier level is determined based on the particular tier level number and expressed by the formula:

(31) $\% \text{of length of base level perimeter} = 1 - (0.2 + .15 * (\text{tier level number}))$, including $\pm 5\%$. 10. The system of example 7, wherein the percentages of the base level perimeter length for the tier levels ranges from 15-70%. 11. The system of example 10, wherein the percentage of the base level perimeter length is selected from the group consisting of 60-70%, 45-55%, 30-40%, and 15-25%. 12. The system of example 10, wherein the percentage of the base level perimeter length for the first tier level is 60-70%. 13. The system of example 1, wherein the base pan further comprises one or more compressor structural supports and the distance from a center of any of the one or more compressors to the closest compressor structural support is equal to two times the smallest bolt circle diameter or less. 14. The system of example 1, wherein the base pan further comprises one or more compressor structural supports and the distance from a center of any of the one or more compressors to the closest compressor structural support is equal to the smallest bolt circle diameter or less. 15. The system of example 1, wherein the vertical forces of the one or more compressors are evenly distributed on the base pan. 16. The system of example 1, further comprising an outdoor section including the one or more compressors and one or more outdoor heat exchangers and an indoor section including the one or more indoor heat exchangers, and wherein the base pan only supports the outdoor section. 17. The system of example 1, further comprising an outdoor section including the one or more compressors and one or more outdoor heat exchangers and an indoor section including the one or more indoor heat exchangers, and wherein the base pan supports the outdoor section and the indoor section. 18. A base pan for a heating, ventilation, and air conditioning (HVAC) system comprising one or more compressors to be mounted on the base pan, the base pan comprising a base level and one or more tier levels higher than the base level, wherein a distance from any point on a perimeter of the first tier level to a nearest point on a perimeter of the base level is based on the smallest bolt circle diameter of the one or more compressors and the tier level perimeter lengths are based on a percentage of the base level perimeter length. 19. The base pan of example 18, wherein the distance from any point on the perimeter of the first tier level to the nearest point on the perimeter of the base level is equal to two times the smallest bolt circle diameter or less. 20. The base pan of example 19, wherein the distance from any point on the perimeter of the first tier level to the nearest point on the perimeter of the base level is equal to the smallest bolt circle diameter or less. 21. The base pan of example 18, wherein the base pan further comprises more than one tier level and the distance from any point on the perimeter of each tier level and the nearest point on the perimeter of the adjacent tier level is based on the smallest bolt circle diameter. 22. The system of example 21, wherein the distance from any point on the perimeter of each tier level and the nearest point on the perimeter of the adjacent tier level is equal to the smallest bolt circle of the compressor or less. 23. The system of example 21, wherein the distance from any point on the perimeter of each tier level and the nearest point on the perimeter of the adjacent tier level is equal to half the smallest bolt circle of the compressor or less. 24. The base pan of example 18, wherein the base pan further comprises more than one tier level and each tier level perimeter length is based on a percentage of the base level perimeter length. 25. The base pan of example 24, wherein the percentage of the base level perimeter length for each tier level perimeter length is selected based on the tier level number. 26.

The system of example 25, wherein the percentage of the base level perimeter length of each tier level is determined based on the particular tier level number and expressed by the formula:
$$(32) \text{ \%oflengthofbaselevelperimter} = 1 - (0.2 + .15 * (\text{tierlevelnumber})), \text{ including } \pm 5\% .$$
 27. The base pan of example 24, wherein the percentages of the base level perimeter length for the tier levels ranges from 15-70%. 28. The base pan of example 27, wherein the percentage of the base level perimeter length is selected from the group consisting of 60-70%, 45-55%, 30-40%, and 15-25%. 29. The base pan of example 27, wherein the percentage of the base level perimeter length for the first tier level is 60-70%. 30. The base pan of example 18, wherein the base pan further comprises one or more compressor structural supports positioned such that the distance from the compressor structural support to a center of the compressor to be mounted on the compressor structural support is equal to two times the smallest bolt circle diameter or less. 31. The base pan of example 18, wherein the base pan further comprises one or more compressor structural supports positioned such that the distance from the compressor structural support to a center of the compressor to be mounted on the compressor structural support is equal to the smallest bolt circle diameter or less. 32. The base pan of example 18, wherein the base pan is sized and shaped such that the vertical forces of the one or more compressors are evenly distributed on the base pan when mounted. 33. The base pan of example 18, wherein the HVAC system further comprises an outdoor section including the one or more compressors and an indoor section comprising one or more indoor heat exchangers, and wherein the base pan only supports the outdoor section. 34. The base pan of example 18, wherein the HVAC system further comprises an outdoor section including the one or more compressors and an indoor section comprising one or more indoor heat exchangers, and wherein the base pan supports the outdoor section and the indoor section.

(33) Certain terms are used throughout the description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not function.

(34) The embodiments disclosed should not be interpreted, or otherwise used, as limiting the scope of the disclosure, including the claims. It is to be fully recognized that the different teachings of the embodiments discussed may be employed separately or in any suitable combination to produce desired results. In addition, one skilled in the art will understand that the description has broad application, and the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to suggest that the scope of the disclosure, including the claims, is limited to that embodiment.

Claims

1. A heating, ventilation, and air conditioning (HVAC) system comprising: one or more compressors; one or more outdoor heat exchangers; one or more indoor heat exchangers; and a base pan configured for mounting one or more compressors thereon and comprising a base level and one or more tier levels higher than the base level, wherein a distance from any point on a perimeter of the first tier level to a nearest point on a perimeter of the base level is based on the smallest bolt circle diameter of the one or more compressors and the tier level perimeter lengths are based on a percentage of the base level perimeter length.
2. The system of claim 1, wherein the distance from any point on the perimeter of the first tier level to the nearest point on the perimeter of the base level is equal to two times the smallest bolt circle diameter or less.
3. The system of claim 2, wherein the distance from any point on the perimeter of the first tier level to the nearest point on the perimeter of the base level is equal to the smallest bolt circle diameter or less.
4. The system of claim 1, wherein the base pan further comprises more than one tier level and the

distance from any point on the perimeter of each tier level and the nearest point on the perimeter of an adjacent tier level is based on the smallest bolt circle diameter.

5. The system of claim 4, wherein the distance from any point on the perimeter of each tier level and the nearest point on the perimeter of an adjacent tier level is equal to the smallest bolt circle of the compressor or less.

6. The system of claim 4, wherein the distance from any point on the perimeter of each tier level and the nearest point on the perimeter of an adjacent tier level is equal to half the smallest bolt circle of the compressor or less.

7. The system of claim 1, wherein the base pan further comprises more than one tier level and each tier level perimeter length is based on a percentage of the base level perimeter length.

8. The system of claim 7, wherein the percentage of the base level perimeter length for each tier level perimeter length is selected based on the tier level number.

9. The system of claim 8, wherein the percentage of the base level perimeter length of each tier level is determined based on the particular tier level number and expressed by:

$\% \text{of length of base level perimeter} = 1 - (0.2 + .15 * (\text{tier level number})), \text{ including } \pm 5\%.$

10. The system of claim 7, wherein the percentages of the base level perimeter length for the tier levels ranges from 15-70%.

11. The system of claim 10, wherein the percentage of the base level perimeter length is selected from the group consisting of 60-70%, 45-55%, 30-40%, and 15-25%.

12. The system of claim 10, wherein the percentage of the base level perimeter length for the first tier level is 60-70%.

13. The system of claim 1, wherein the base pan further comprises one or more compressor structural supports and the distance from a center of any of the one or more compressors to the closest compressor structural support is equal to two times the smallest bolt circle diameter or less.

14. The system of claim 1, wherein the base pan further comprises one or more structural supports and the distance from a center of any of the one or more compressors to the closest structural support is equal to the smallest bolt circle diameter or less.

15. The system of claim 1, wherein vertical forces of the one or more compressors are evenly distributed on the base pan.

16. The system of claim 1, further comprising an outdoor section including the one or more compressors and one or more outdoor heat exchangers and an indoor section including the one or more indoor heat exchangers, and wherein the base pan only supports the outdoor section.

17. The system of claim 1, further comprising an outdoor section including the one or more compressors and one or more outdoor heat exchangers and an indoor section including the one or more indoor heat exchangers, and wherein the base pan supports the outdoor section and the indoor section.

18. A base pan for a heating, ventilation, and air conditioning (HVAC) system, wherein the base pan is configured for mounting one or more compressors thereon, and the base pan comprising a base level and one or more tier levels higher than the base level, wherein a distance from any point on a perimeter of the first tier level to a nearest point on a perimeter of the base level is based on the smallest bolt circle diameter of the one or more compressors and the tier level perimeter lengths are based on a percentage of the base level perimeter length.

19. The base pan of claim 18, wherein the distance from any point on the perimeter of the first tier level to the nearest point on the perimeter of the base level is equal to two times the smallest bolt circle diameter or less.

20. A method for manufacturing a base pan for mounting one or more compressors of a heating, ventilation, and air-conditioning ("HVAC"), the method comprising: configuring the base pan to comprise a base level and one or more tier levels higher than the base level; determining a distance from any point on a perimeter of the first tier level to a nearest point on a perimeter of the base

level based on the smallest bolt circle diameter of the one or more compressors; and determining the tier level perimeter lengths based on a percentage of the base level perimeter length.
